

CCA-BDP Reviewer Summary

Methods, Results, Caveats, and Reproducibility - Generated 2026-07-06 16:10

Methods Summary

Random Forest classifier (500 trees, class_weight="balanced"), trained on 44 TCGA-CHOL samples using a 20-gene clinical biomarker panel selected for reproducibility across independent GEO cohorts. Externally validated on 3 independent GEO cohorts (444 pooled samples). Explained via SHAP (TreeExplainer, tree_path_dependent) and permutation importance. Biological validation via GO/KEGG/Reactome/WikiPathways (Enrichr) and STRING PPI. Survival analysis via Kaplan-Meier, univariate Cox regression, and a landmark-time discrimination method. Drug target discovery via Open Targets and DGIdb.

Results Summary

External ROC-AUC=0.933 (Sensitivity=0.111, Specificity=0.986). Top biomarker by SHAP: USH2A. Top drug repurposing candidate: DGAT2 -> ERVOGASTAT.

Statistical Caveats & Limitations

GSE32225 and GSE89749 predict entirely class 0 - sensitivity evidence is effectively carried by GSE26566 alone. Survival analysis (n=34, 18 events) is exploratory; Kaplan-Meier and continuous Cox regression disagree on which gene is prognostic, an expected consequence of small-sample method sensitivity, not an error. No pathway enrichment beyond GO Molecular Function survives multiple-testing correction. Permutation importance on external data diverges substantially from training-based SHAP/RF importance, reflecting real distribution shift.

Reproducibility Information

Model trained 2026-07-04 20:50:19, scikit-learn 1.8.0, Python 3.11.15, random_state=42 throughout. All intermediate artifacts (trained model, feature order, per-module CSV/JSON outputs, 300 DPI figures) are version-controlled and regeneratable by re-running each module's orchestrator script.